

# MINI ASSESSMENT

Python Rule-Based Conversational Chatbot

|              |                                 |
|--------------|---------------------------------|
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## 01 AIM

To design and develop an interactive, rule-based conversational chatbot in Python using a continuous **while True** loop, **input()**, **.lower().strip()**, and **if-elif-else** conditional branching. The chatbot asks the user's name first, recognizes flexible greetings and multiple topics, uses the user's name in a response, handles unknown messages, and exits when the user types **bye** or **exit**.

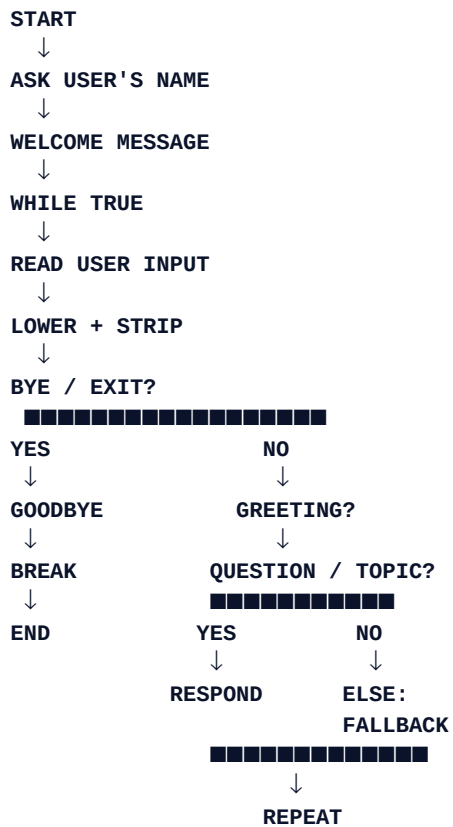
## 02 PROCEDURE

1. Ask the user's name first and clean it with **.strip()**.
2. Display a welcome message using the user's name.
3. Start continuous conversation using **while True**.
4. Receive messages with **input()** and clean them using **.lower().strip()**.
5. Use **bye** or **exit** as termination commands and stop with **break**.
6. Detect greetings such as **hello**, **hi**, and **hey** using the **in** operator.
7. Handle at least three questions/topics.
8. Use the user's name in at least one response.
9. Use **else** to provide a useful fallback for unknown messages.

## 03 TARGET FLOW

| INPUT        | PROCESS         | RESPOND          | REPEAT     | EXIT               |
|--------------|-----------------|------------------|------------|--------------------|
| User message | lower().strip() | Rule-based reply | while True | bye / exit + break |

## 04 FLOWCHART



## 05 PYTHON CODE

```
def rule_based_bot():
    # Ask the user's name first
    name = input("Bot: Hi! What is your name? ").strip()

    print(f"Bot: Nice to meet you, {name}!")
    print("Bot: Ask me something or type 'bye' or 'exit' to quit.\n")

    while True:
        user_input = input("You: ").lower().strip()

        if user_input in ["bye", "exit"]:
            print(f"Bot: Goodbye, {name}! Have a great day!")
            break

        elif "hello" in user_input or "hi" in user_input or "hey" in user_input:
            print(f"Bot: Hello {name}! How can I help you?")

        elif "how are you" in user_input:
            print("Bot: I am doing great, thank you!")

        elif "your name" in user_input:
            print("Bot: I am a simple Python rule-based chatbot.")

        elif "what is python" in user_input or "python" in user_input:
            print("Bot: Python is a popular programming language.")

        elif "help" in user_input:
            print("Bot: You can greet me, ask about Python, "
                  "ask my name, or ask how I am.")

        else:
            print(f"Bot: Sorry {name}, I don't understand that yet. "
                  "Try asking about Python or say hello.")

if __name__ == "__main__":
    rule_based_bot()
```

## 06 RESULT

| INPUT              | OUTPUT                                                                            |
|--------------------|-----------------------------------------------------------------------------------|
| Name: Vardhan      | Nice to meet you, Vardhan!                                                        |
| hello bot          | Hello Vardhan! How can I help you?                                                |
| how are you        | I am doing great, thank you!                                                      |
| what is python     | Python is a popular programming language.                                         |
| what is your name? | I am a simple Python rule-based chatbot.                                          |
| can you dance?     | Sorry Vardhan, I don't understand that yet. Try asking about Python or say hello. |
| bye                | Goodbye, Vardhan! Have a great day!                                               |

## 07 REQUIREMENT CHECKLIST

| REQUIREMENT               | STATUS             |
|---------------------------|--------------------|
| Ask user's name first     | ✓ Completed        |
| while True                | ✓ Completed        |
| input()                   | ✓ Completed        |
| lower().strip()           | ✓ Completed        |
| Two or more greetings     | ✓ hello / hi / hey |
| Three or more topics      | ✓ 4 topics         |
| if / elif / else          | ✓ Completed        |
| bye / exit + break        | ✓ Completed        |
| Use user's name           | ✓ Completed        |
| Unknown-message fallback  | ✓ Completed        |
| Flexible matching with in | ✓ Completed        |

## 08 CONCLUSION

The rule-based conversational chatbot was successfully implemented using Python conditional branching and a continuous conversation loop. It follows the required INPUT → PROCESS → RESPOND → REPEAT → EXIT flow, supports flexible greeting detection, handles multiple topics, uses the user's name, and provides a useful fallback response for unknown messages. The program terminates gracefully using **break** when the user types **bye** or **exit**.